CHE 4990/CHE5990 **Fall 2026**

Reading Assignment 4 Questions (40 points)

Due 11.59 PM on September 23, 2026

Reading article:

The reading article for this assignment is

https://www.eia.gov/energyexplained/oil-and-petroleum-products/refining-crude-oil-the-refining-process.php

Please read all four tabs in this article before answering the questions.

Question 1 – based on the class lecture (26 points)

An ideal Rankine cycle operates between a boiler pressure of 8 MPa and a condenser pressure of 10 kPa. Steam enters the turbine at 480°C. The mass flow rate of steam is 75 kg/s. Assume the pump and turbine are isentropic. Using steam tables, determine:

a. h and s at each of the four states (18 points).
b. Pump work (4 points).
c. Boiler heat input (4 points).

Steam tables at https://pages.mtu.edu/~tbco/cm3230/steamtables.pdf

Question 2 – based on the reading article (4 points)

Why can a refinery produce more than 42 gallons of products from a 42-gallon barrel of crude oil, and explain why this does not violate conservation of mass.

Question 3 – based on the reading article (5 x 2 = 10 points)

You are an engineer at a refinery that primarily produces gasoline, diesel, and jet fuel. You have a choice between two crude-oil feedstocks:
Crude A: light crude, relatively low sulfur content.
Crude B: heavy crude, relatively high sulfur content.
Assume both can be purchased and delivered to the refinery. Based only on information and engineering implications supported by the reading article, answer the following:

a. Which crude would generally require more extensive processing to produce gasoline, diesel, and jet fuel? Explain why.
b. Would you expect the two crude oils to produce the same distribution of products following simple distillation?